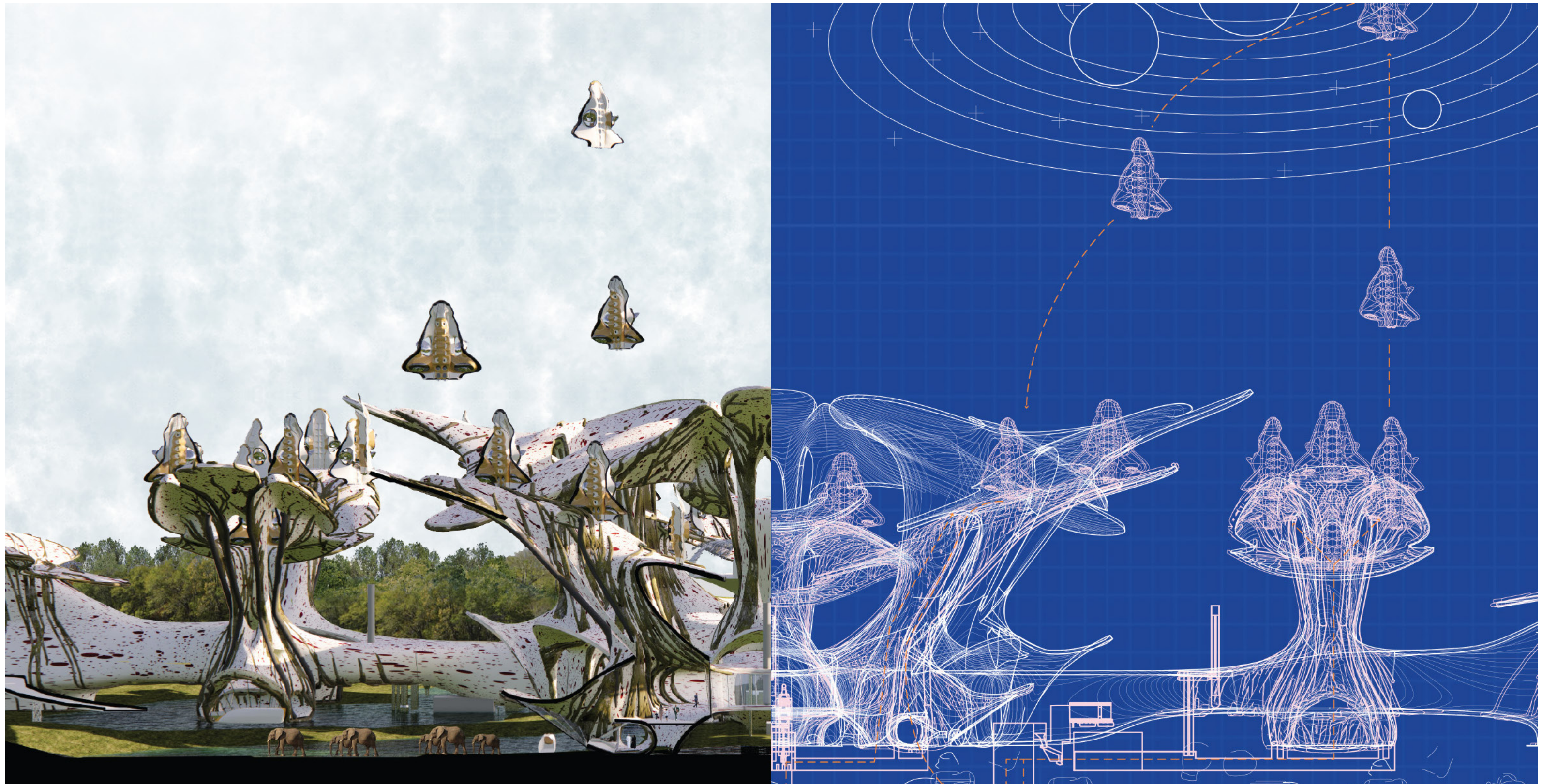




02. Astra(Fentress Global Challenge)

Competition: Fentress 2100 Global Airport Challenge
Partner: Jonathan Liang
Award: Top 10 Finalists

This architectural competition project envisions a **next-generation airport for the year 2100**, merging **sustainability with computational design innovation**. Developed using **Python scripting and the Grasshopper** plugin in Rhino, the design explores creative, performance-driven forms informed by **environmental data, material efficiency, and circular economy principles**.

Central to the concept is a closed-loop system that recycles urban waste and converts it into clean energy to power the airport, dramatically **reducing its carbon footprint and reliance on fossil fuels**. The design also anticipates the rise of **vertical takeoff and landing (VTOL) aircraft**, which require less runway space and allow for a more compact, multi-level terminal layout

that optimizes land use. Additionally, the project **reclaims adjacent landfill sites, transforming them into thriving wildlife refuges** that restore local ecosystems, mitigate soil contamination, and support native species. A responsive skin system, optimized through **environmental simulation, regulates daylight, heat gain, and airflow across the structure**. Blending high-tech

infrastructure with regenerative land use, the proposal redefines how airports can operate as ecologically integrated systems that serve both people and the planet while adapting to **future transportation and climate needs**.